LaTeX Source Code: Q.1 to Q.20

Q.1

```
\documentclass{article}
\usepackage{amsmath,amssymb}
\begin{document}
        \[
        A=\begin{bmatrix}
                x & \dfrac{x}{y} & 0\\[3mm]
                e^{-x} & -\tfrac{1}{y} & \sqrt[3]{x}\\
                \sin\theta & -3 & 2
        \end{bmatrix}\in\mathbb{R}^{3\times 3}.
        \]
\end{document}
```

Q.2

```
\documentclass{article}
\usepackage{array,multirow}
\usepackage{comment}
\begin{document}
        \begin{center}
                \begin{tabular}{|c|*{4}{m{12mm}|}}
                        \hline
                        \multicolumn{5}{|c|}{Sales Data}\\ \hline
                        \multirow{2}{*}{Seasons} & \multicolumn{4}{c|}{Salesman}\\ \cline{2-5}
                        & A & B & C & D\\ \hline
                        Summer & 36 & 21 & 35 & 35\\ \hline
                        Winter & 28 & 29 & 31 & 32\\ \hline
                        Monsoon & 26 & 28 & 29 & 29\\ \hline
                \end{tabular}
        \end{center}
\end{document}

\begin{comment}


\end{comment}
```

Q.3

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
        The arc length of the curve $y=f(x)$ over the interval $a\leq x\leq b$ is
        $\displaystyle\int_a^b\sqrt{1+[f'(x)]^2}\,dx$. For the function
        $f(x)=x^3,\ 0\leq x\leq 1$, find the approximate arc length correct to four
        decimal places using Trapezoidal rule with $h=0.1$.
\end{document}
```

Q.4

```
\documentclass{article}
\usepackage{xcolor,enumerate}
\begin{document}
        \begin{enumerate}[(a)]
                \item {\bfseries\Large\textcolor{red}{\underline{Descriptive Research}}}
                \item {\bfseries\large\textcolor{violet}{\underline{Descriptive Research}}}
                \item {\bfseries\tiny\textcolor{blue}{Descriptive Research}}
                \item {\bfseries\huge\textcolor{brown}{Descriptive Research}}
                \item {\bfseries\Huge\textcolor{green}{\textit{Descriptive Research}}}
        \end{enumerate}
\end{document}
```

Q.5

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
        \[
        f(x)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\left(\frac{n\pi x}{l}\right)+b_n\sin\left(\frac{n\pi x}{l}\right)\
    right).
        \]
\end{document}
```

Q.6

```
\documentclass{article}
\usepackage{amsmath}
\begin{document}
        \[
        f(t)=\begin{cases}
                0, & t<0,\\
                e^{-t}, & t>0.
        \end{cases}
        \]
\end{document}
```

Q.7

```
\documentclass{exam}
\usepackage{amsmath}
\renewcommand\questionlabel{Q.\thequestion}
\begin{document}
        \begin{center}
                \textbf{Demo Question Paper\\Differential Equations}
        \end{center}
        \hrule\vspace{2mm}

        \begin{questions}
                \setcounter{question}{1}
                \question
                \begin{parts}
                        \part Use suitable Fourier transform to solve the following partial differential equation:
                        \begin{equation*}
                                \begin{array}{l}
                                        \dfrac{\partial^2u}{\partial x^2}=\dfrac14\dfrac{\partial^2u}{\partial t^2},\
    quad -\infty<x<\infty,\ t>0,\\[3mm]
                                        u(x,t)\to 0\ \text{as}\ |x|\to\infty,\\[2mm]
                                        u(x,0)=0,\quad u_t(x,0)=e^{-|x|},\quad -\infty<x<\infty.
                                \end{array}
                        \end{equation*}
                \end{parts}
        \end{questions}
\end{document}
```

Q.8

```
\documentclass{article}
\usepackage{tikz,amsmath}
\usepackage{comment}
\begin{document}

\noindent\textbf{Q.8.} Use Liebmann's iteration formula to solve Laplace equation
$\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=0$ at the interior mesh points of the square
    region with boundary values given in the following figure.

\begin{center}
\begin{tikzpicture}[scale=1.5]
\draw (0,0) grid (4,4);
\foreach \n/\x/\y in {1/1/1,2/2/1,3/3/1,4/1/2,5/2/2,6/3/2,7/1/3,8/2/3,9/3/3}
  \node[circle,draw,inner sep=2pt,label={above right:$u_{\n}$}] at (\x,\y) {};
\foreach \v [count=\i] in {50,100,50}{\node[above] at (\i,4) {\v}; \node[below] at (\i,0) {\v};}
\foreach \v [count=\i] in {100,200,100}{\node[left] at (0,\i) {\v}; \node[right] at (4,\i) {\v};}
\node[left] at (0,0) {0}; \node[left] at (0,4) {0};
\node[right] at (4,0) {0}; \node[right] at (4,4) {0};
\end{tikzpicture}
\end{center}

\noindent[Take 2 iterations and obtain the result correct up to three decimal places.]
\begin{comment}


        multirow provides \multirow{2}{*}{Seasons}, which makes the \"Seasons cell span two rows. The * means the cell
    takes its natural width.
        array provides the m{width} column type, a fixed-width column whose content is vertically centred. It also
    provides *{n}{...} for repeating column specs. Many LaTeX distributions load it automatically when tabular is used,
     but 'its safest to include it.

        Commands built into LaTeX (no package needed):
        \multicolumn{5}{|c|}{Sales Data} merges 5 columns into one centred cell, and the | keeps the borders.
        \hline draws a full-width horizontal line.
        \cline{2-5} draws a line only under columns 2 to 5, so it leaves the \"Seasons cell open.

        --------------------
\end{comment}
\end{document}
```

Q.9

```
\documentclass{beamer}
\usetheme{Madrid}
\title{Quantum Mechanics}
\subtitle{Motivation in Research}
\author{Nikhil S Sulghur}
\institute{Mumbai University}
\date{August 15, 2026}

\begin{document}

        \begin{frame}\titlepage\end{frame}

        \begin{frame}{Axioms of Quantum Mechanics}
                \begin{enumerate}
                        \item \textbf{State:} A system is described by a normalized state vector $|\psi\rangle$ in a Hilbert space.
                        \item \textbf{Observables:} Every measurable quantity corresponds to a Hermitian operator $\hat A$.
                        \item \textbf{Measurement:} The only possible results are the eigenvalues $a_n$ of $\hat A$, with probability $|\langle a_n|\psi\rangle|^2$.
                        \item \textbf{Collapse:} After measuring $a_n$, the state becomes the eigenstate $|a_n\rangle$.
                        \item \textbf{Evolution:} $i\hbar\,\dfrac{\partial}{\partial t}|\psi\rangle=\hat H|\psi\rangle$.
                \end{enumerate}
        \end{frame}

        \begin{frame}{References}
                \begin{thebibliography}{1}
                        \bibitem{G} D. J. Griffiths, \emph{Introduction to Quantum Mechanics}, Pearson.
                \end{thebibliography}
        \end{frame}

        \begin{frame}
                \centering
                {\Huge Thank You}\\[5mm]
                I thank the Department of Mathematics for this opportunity.
        \end{frame}

\end{document}
```

Q.10

```latex
\documentclass{article}
\usepackage{xcolor}
\begin{document}

        %% ======================================================================
        %% (a) TEXT STYLE AND EMPHASIS
        %% These commands change how characters look (shape, weight, family, colour).
        %% Each has two forms:
        %%   - command form  \textbf{text}  : affects only the text in the braces (preferred)
        %%   - switch form   {\bfseries text} : affects everything after it, inside the group
        %%
        %%  \textbf{..}  / \bfseries  : bold weight, used for headings and key terms.
        %%  \textit{..}  / \itshape   : italic shape, used for titles of books, foreign words.
        %%  \textsl{..}  / \slshape   : slanted shape (upright letters tilted).
        %%  \textsc{..}  / \scshape   : SMALL CAPS, used for names and abbreviations.
        %%  \texttt{..}  / \ttfamily  : typewriter (monospace) family, used for code.
        %%  \textsf{..}  / \sffamily  : sans-serif family.
        %%  \underline{..}            : underlines the text.
        %%  \emph{..}                 : logical emphasis. Normally gives italics, but inside
        %%                              italic text it switches back to upright, so it is
        %%                              safer than \textit for stressing a word.
        %%  \textcolor{colour}{..}    : coloured text; needs the xcolor package.
        %% Styles can be combined by nesting, e.g. \textbf{\textit{bold italic}}.
        %% ======================================================================
        \section*{(a) Text style and emphasis}
        \textbf{Bold}, \textit{italic}, \textsc{small caps}, \texttt{typewriter}, \underline{underlined}, \emph{
    emphasised}, \textcolor{red}{red}.

        %% ======================================================================
        %% (b) TEXT SIZING (smallest to largest)
        %% LaTeX has ten size commands. Their actual point size depends on the document
        %% class and its base size option (10pt, 11pt, 12pt). In order:
        %%   \tiny  <  \scriptsize  <  \footnotesize  <  \small  <  \normalsize
        %%   <  \large  <  \Large  <  \LARGE  <  \huge  <  \Huge
        %% \normalsize is the default body size.
        %% These are switches: they change all text that follows. To limit the effect,
        %% put the command and the text together inside braces { }, or inside an
        %% environment, so the size returns to normal at the closing brace.
        %% ======================================================================
        \section*{(b) Text sizing}
        {\tiny tiny} {\scriptsize scriptsize} {\footnotesize footnotesize} {\small small} {\normalsize normalsize} {\
    large large} {\Large Large} {\LARGE LARGE} {\huge huge} {\Huge Huge}

        %% ======================================================================
        %% (c) PARAGRAPH ALIGNMENT AND ENVIRONMENTS
        %% By default LaTeX justifies text (both margins straight). An environment
        %% \begin{name} ... \end{name} applies a setting to the text inside it only.
        %%   center      : centres every line; used for titles, figures, tables.
        %%   flushleft   : aligns to the left margin, right edge ragged.
        %%   flushright  : aligns to the right margin, left edge ragged.
        %%   quote       : indented block for short quotations.
        %%   quotation   : indented block for longer quotations, with paragraph indents.
        %%   verbatim    : prints the text exactly as typed, in typewriter font; used for code.
        %% Paragraph and line breaks:
        %%   blank line or \par : starts a new paragraph (with first-line indent).
        %%   \\ or \newline     : breaks the line without starting a new paragraph.
        %%   \noindent          : removes the indent of the next paragraph.
        %% Inline switches \centering, \raggedright, \raggedleft do the same as the
        %% environments but are used inside other structures like tables and boxes.
        %% ======================================================================
        \section*{(c) Paragraph alignment and environments}
        \begin{center} This line is centred. \end{center}
        \begin{flushleft} This line is left aligned. \end{flushleft}
        \begin{flushright} This line is right aligned. \end{flushright}

        %% ======================================================================
        %% (d) DOCUMENT STRUCTURE
        %% Every LaTeX file has a preamble (before \begin{document}) and a body.
                %%   \documentclass{..}   : first line; sets the type: article, report, book,
                %%                          letter, beamer. Options go in [ ], e.g. [12pt,a4paper].
                %%   \usepackage{..}      : loads extra features (amsmath, tikz, xcolor, ...).
                %%   \title{..} \author{..} \date{..} then \maketitle : build the title block.
                %%   \begin{document} ... \end{document} : everything printed goes between these.
                %% Sectioning commands divide the text and number it automatically:
                %%   \part, \chapter (report and book only), \section, \subsection,
                %%   \subsubsection, \paragraph.
                %%   Adding a star (\section*{..}) removes the number and the contents entry.
```

```
                %% Other structural tools:
                %%   \tableofcontents     : builds the contents page; compile twice.
                %%   \begin{abstract}..   : short summary at the start (article, report).
                        %%   \label{..} and \ref{..} : cross-refer to a numbered section, equation or figure.
                        %% =========================================================================
                        \section*{(d) Document structure}
                        \subsection*{A subsection}
                        \subsubsection*{A subsubsection}
                        This page uses \verb|\documentclass|, \verb|\usepackage|, \verb|\section*| and the \verb|
    document| environment.

                        %% =========================================================================
                        %% (e) LISTS AND LAYOUT
                        %% Lists (each entry begins with \item; lists can be nested):
                        %%   itemize     : bulleted list.
                        %%   enumerate   : numbered list (1., 2., ...).
                        %%   description : list of terms in bold, each followed by its meaning: \item[term].
                        %% Layout commands:
                        %%   \newpage, \clearpage : start a new page.
                        %%   \vspace{..}, \hspace{..} : insert vertical or horizontal space.
                        %%   \hrule               : draws a horizontal line.
                        %%   \hfill               : stretchable space; pushes text to the right.
                        %%   \begin{minipage}{width}..\end{minipage} : box of fixed width.
                        %%   \begin{multicols}{n} : n text columns (needs the multicol package).
                                %%   \usepackage{geometry} : sets page margins and size.
                                %% =============================================================================
                                \section*{(e) Lists and layout}
                                \begin{itemize} \item Apple \item Mango \end{itemize}
                                \begin{enumerate} \item First \item Second \end{enumerate}
                                %\begin{description} \item[LaTeX] A document preparation system. \end{description}
                                \vspace{5mm}\hrule

                        \end{document}
```

Q.11

```
\documentclass{article}
\usepackage{fancyhdr}

%% ========================================================================
%% USE OF THE fancyhdr PACKAGE
%% By default LaTeX prints only a page number (in the footer or header,
%% depending on the class). The fancyhdr package lets you control what is
%% printed in the header (top of page) and footer (bottom of page).
%%
%% Each of header and footer has three positions: Left, Centre, Right.
%% Commands (used inside \fancyhf, \fancyhead, \fancyfoot):
%%   \pagestyle{fancy}   : turns on the fancyhdr style for the whole document.
%%   \fancyhf{}          : clears all existing header and footer fields.
%%                         Always use it first, to start from a blank layout.
%%   \fancyhead[L]{..}   : left header      \fancyhead[C]{..} : centre header
%%   \fancyhead[R]{..}   : right header
%%   \fancyfoot[L]{..}   : left footer      \fancyfoot[C]{..} : centre footer
%%   \fancyfoot[R]{..}   : right footer
%%   \thepage            : prints the current page number.
%%   \leftmark / \rightmark : print the current chapter / section title.
%%   \renewcommand{\headrulewidth}{0.4pt} : thickness of the line under the
%%                          header (0pt removes it).
%%   \renewcommand{\footrulewidth}{0.4pt} : thickness of the line above the
%%                          footer (0pt by default, so no line).
%%   \setlength{\headheight}{14pt} : height reserved for the header; set it
%%                          if LaTeX warns that the header is too small.
%% For two-sided documents (book/report with twoside), use E and O in the
%% brackets for even and odd pages, e.g. \fancyhead[LE,RO]{..}.
%% ========================================================================

\pagestyle{fancy}
\fancyhf{}                                  % clear everything first
\fancyhead[L]{Research Methodology}         % header, left
\fancyhead[R]{Nikhil S Sulghur}             % header, right
\fancyfoot[L]{Mumbai University}            % footer, left
\fancyfoot[R]{Page \thepage}                % footer, right
\renewcommand{\headrulewidth}{0.4pt}        % line under header
\renewcommand{\footrulewidth}{0.4pt}        % line above footer
\setlength{\headheight}{14pt}

\begin{document}
        \section{Introduction}
        This is the first page. The header and footer set above appear on every page.
        \newpage
        \section{Next Section}
        This is the second page, showing the page number change in the footer.
\end{document}
```

Q.12

```
\documentclass{letter}
\setlength{\longindentation}{0pt}
\address{\parbox{1\textwidth}{Nikhil S Sulghur\\M.Sc.-I, Department of Mathematics\\Mumbai University}}
\signature{Nikhil S Sulghur}
\begin{document}
        \begin{letter}{The Head of the Department\\Department of Mathematics\\Mumbai University}
                \opening{Respected Sir/Madam,}

                \textbf{Subject: Request for permission to use MATLAB}

                I am a student of M.Sc.-I, Department of Mathematics. I kindly request your permission to use MATLAB in the departmental computer laboratory for my research project work.

                I shall use the software only for academic purposes and will follow all laboratory rules. I would be grateful for your kind approval.

                \closing{Yours faithfully,}
        \end{letter}
\end{document}
```

Q.13

```
\documentclass[12pt]{report}
\usepackage{makeidx}
\makeindex
\newcommand{\txt}[1]{#1 This topic is discussed in Rudin~\cite{R} and forms a basic part of real analysis.}

\title{Research in Mathematics:\\Real Analysis}
\author{Nikhil S Sulghur}
\date{\today}

\begin{document}
        \maketitle

        \chapter*{Declaration}
        I hereby declare that this project report is my own work and has not been submitted to any other university or institution for any degree or diploma. All sources used are acknowledged in the bibliography.

        \chapter*{Acknowledgement}
        I express my sincere gratitude to the Department of Mathematics, Mumbai University, for giving me this opportunity. I also thank my teachers and friends for their constant guidance and support.

        \tableofcontents

        \chapter{Number Systems}
        \section{Introduction}
        \txt{This chapter introduces the real number system\index{real numbers}.}
        \section{Main Content}
        \txt{We study ordered sets\index{ordered set} and the least upper bound property.}
        \section{Examples}
        \txt{The rational numbers\index{rational numbers} do not have this property.}

        \chapter{Basic Topology}
        \section{Introduction}
        \txt{This chapter introduces metric spaces\index{metric space}.}
        \section{Main Content}
        \txt{We study open sets and compact sets\index{compact set}.}
        \section{Examples}
        \txt{Closed and bounded intervals are compact subsets of the real line.}

        \chapter{Sequences and Series}
        \section{Introduction}
        \txt{This chapter introduces numerical sequences\index{sequence}.}
        \section{Main Content}
        \txt{We study convergence and Cauchy sequences\index{Cauchy sequence}.}
        \section{Examples}
        \txt{The sequence $1/n$ converges to zero.}

        \chapter{Continuity}
        \section{Introduction}
        \txt{This chapter introduces limits of functions\index{limit}.}
        \section{Main Content}
        \txt{We study continuous functions\index{continuity} on metric spaces.}
        \section{Examples}
        \txt{Polynomial functions are continuous on the whole real line.}

        \chapter{Conclusion}
        In this report we studied the basic ideas of real analysis, namely number systems, topology, sequences and continuity, following Rudin~\cite{R}. These ideas form the foundation for further study of differentiation and integration.

        \begin{thebibliography}{9}
                \bibitem{R} W. Rudin, \emph{Principles of Mathematical Analysis}, 3rd ed., McGraw-Hill, 1976.
        \end{thebibliography}

        \printindex
\end{document}
```

Q.14

```
\documentclass{article}
\usepackage{amsmath}
\title{Existence and Uniqueness of Solutions of the Second Order Differential Equations}
\author{Nikhil S Sulghur\\Mumbai University}
\date{\today}

\begin{document}
        \maketitle

        \begin{abstract}
                We study the existence and uniqueness of solutions of second order differential equations. We state the Picard--Lindel\"of type result for the initial value problem and give a simple example.
        \end{abstract}

        \section{Introduction}
        Second order differential equations arise in many problems of physics and engineering. A basic question is whether a solution exists and whether it is unique; see Tsamatos and Ntouyas \cite{TN} for boundary value problems of this type.

        \section{Preliminaries}
        Consider the initial value problem
        \[
        y''=f(x,y,y'),\quad y(x_0)=y_0,\quad y'(x_0)=y_1.
        \]
        Setting $z=y'$ turns it into a first order system, so results for systems can be used.

        \section{Main Result}
        \textbf{Theorem.} If $f$ is continuous and Lipschitz continuous in $y$ and $y'$ on a rectangle around $(x_0,y_0, y_1)$, then the problem has a unique solution on some interval around $x_0$.

        \section{Example}
        For $y''+y=0$ with $y(0)=0$ and $y'(0)=1$, the function $f$ is linear, so the theorem applies. The unique solution is $y=\sin x$.

        \section{Conclusion}
        Under a Lipschitz condition the initial value problem for a second order equation has exactly one solution. This idea extends to more general boundary value problems.

        \begin{thebibliography}{9}
                \bibitem{TN} P. Ch. Tsamatos and S. K. Ntouyas, Existence and uniqueness of solutions of a general type boundary-value problem for 2nd-order functional-differential equations, \emph{J. Math. Anal. Appl.}, Elsevier, 1993.
        \end{thebibliography}
\end{document}
```

Q.15

```
\documentclass{article}
\usepackage{fancyhdr}

%% ========================================================================
%% 1. DISPLAYING THE PAGE NUMBER IN THE HEADER OR FOOTER
%% LaTeX stores the current page number in a counter called "page".
%% The command \thepage prints its value, so it can be placed anywhere.
%%
%% (a) Built-in page styles (set with \pagestyle{..} for the whole document):
%%     plain       : page number centred in the footer (default for article).
%%     headings    : page number and section title in the header.
%%     myheadings  : page number in the header; the heading text is set by hand
%%                   with \markright{..} or \markboth{..}{..}.
%%     empty       : no header and no footer (no page number).
%%   \thispagestyle{..} changes the style of the current page only,
%%   e.g. \thispagestyle{empty} on a title page.
%%
%% (b) Full control with the fancyhdr package:
%%     \pagestyle{fancy}  turns it on, \fancyhf{} clears the default fields.
%%     Then put \thepage in any of the six positions:
%%       \fancyhead[L]{..}  \fancyhead[C]{..}  \fancyhead[R]{..}   (header)
%%       \fancyfoot[L]{..}  \fancyfoot[C]{..}  \fancyfoot[R]{..}   (footer)
%%     e.g. \fancyhead[R]{\thepage}  puts the number at the top right.
%%     \fancyfoot[C]{Page \thepage}  prints "Page n" at the bottom centre.
%%
%% 2. CHANGING THE FORMAT OF THE PAGE NUMBER
%%   \pagenumbering{arabic}  : 1, 2, 3, ...   (default)
%%   \pagenumbering{roman}   : i, ii, iii, ...
%%   \pagenumbering{Roman}   : I, II, III, ...
%%   \pagenumbering{alph}    : a, b, c, ...   \pagenumbering{Alph} : A, B, C, ...
%%
%% 3. RESETTING THE PAGE NUMBER TO A NEW VALUE
%%   (a) \pagenumbering{style} always restarts the numbering from 1, so it is
%%       used when the format changes (e.g. roman preface, then arabic text).
%%   (b) \setcounter{page}{n} sets the page number to any value n. Use it at
%%       the start of the page that should receive the number n (the number
%%       printed on that page becomes n, and later pages continue from n).
%% ========================================================================

\pagestyle{fancy}
\fancyhf{}                          % clear header and footer
\fancyhead[R]{\thepage}             % page number in header, right
\fancyfoot[C]{Page \thepage}        % page number in footer, centre
\setlength{\headheight}{14pt}

\begin{document}
        \pagenumbering{roman}               % front matter: i, ii, ...
        \section*{Preface}
        This page is numbered i.

        \newpage
        \pagenumbering{arabic}              % restarts numbering at 1
        \section{Introduction}
        This page is numbered 1.

        \newpage
        \setcounter{page}{10}               % this page is now numbered 10
        \section{Main Content}
        This page is numbered 10, and the next page will be 11.
\end{document}
```

Q.16

```latex
\documentclass{article}
\usepackage{graphicx}

%% ======================================================================
%% HOW TO INSERT FIGURES IN A LaTeX DOCUMENT
%%
%% 1. LOAD THE PACKAGE
%%    \usepackage{graphicx} in the preamble. It provides \includegraphics.
%%
%% 2. SUPPORTED IMAGE FORMATS
%%    With pdflatex: PNG, JPG/JPEG and PDF. (EPS needs conversion, or latex+dvips.)
%%    Put the image in the same folder as the .tex file (or upload it in Overleaf).
%%
%% 3. THE BASIC COMMAND
%%    \includegraphics[options]{filename}
%%    The extension may be left out. Common options:
%%      width=0.5\textwidth : width as a fraction of the text width (height scales)
%%      height=5cm          : fixed height
%%      scale=0.5           : scale factor
%%      angle=90            : rotate by 90 degrees
%%    Use only one of width/height unless you want the picture distorted.
%%
%% 4. THE figure ENVIRONMENT (floating figure with caption and label)
%%    \begin{figure}[placement] ... \end{figure}
%%    Placement: h = here, t = top of page, b = bottom, p = separate page,
%%               ! = override restrictions. [htbp] tells LaTeX to try them in order.
%%               [H] (float package) forces the exact position.
%%    Inside it:
%%      \centering      : centres the picture
%%      \includegraphics: inserts the picture
%%      \caption{..}    : numbered caption ("Figure 1: ...")
%%      \label{..}      : name for cross-reference; place it AFTER \caption
%%    Refer to the figure in the text with Figure~\ref{fig:name} (compile twice).
%%
%% 5. OTHER WAYS
%%    - Side by side: put two \includegraphics (each width=0.45\textwidth) in
%%      one figure, or use the subcaption package for sub-figures (a), (b).
%%    - \listoffigures prints a list of all figures with page numbers.
%%    - Drawn figures can be made directly with the tikz package.
%% ======================================================================

\begin{document}

        Figure~\ref{fig:sample} shows an inserted picture.

        \begin{figure}[htbp]
                \centering
                \includegraphics[width=0.5\textwidth]{example-image}  % replace with your file, e.g. photo.png
                \caption{A sample figure}
                \label{fig:sample}
        \end{figure}

\end{document}
```

Q.17

```
\documentclass{article}
\begin{document}

        %% ========================================================================
        %% DIFFERENCE BETWEEN THE itemize AND enumerate ENVIRONMENTS
        %%
        %% Both create lists. Each entry begins with \item, and each list is
        %% opened with \begin{..} and closed with \end{..}.
        %%
        %% itemize   : UNORDERED list. Every entry is marked with a bullet symbol
        %%             (a bullet, dash, asterisk, ... depending on nesting level).
        %%             Use it when the order of the items does not matter.
        %%
        %% enumerate : ORDERED list. Entries are numbered automatically
        %%             (1., 2., 3., ...). Use it when the order matters, e.g. steps,
        %%             or when items must be referred to by number.
        %%             The numbers update by themselves when items are added or removed.
        %%
        %% Similarities and further points:
        %%   - Both can be nested up to four levels. The labels change with the level:
        %%       itemize   : bullet, dash, asterisk, centred dot
        %%       enumerate : 1., (a), i., A.
        %%   - The label of any item can be changed by hand with \item[label].
        %%   - Both labels can be customised with the enumitem or enumerate packages,
        %%     e.g. \begin{enumerate}[(a)] gives (a), (b), ...
                %%   - A third list type, description, has bold terms as labels: \item[term].
                %% ========================================================================

                \textbf{Itemize (bullets):}
                \begin{itemize}
                        \item Apple
                        \item Mango
                \end{itemize}

                \textbf{Enumerate (numbers):}
                \begin{enumerate}
                        \item Apple
                        \item Mango
                \end{enumerate}

        \end{document}
```

Q.18

```
\documentclass[12pt]{report}
\newcommand{\blank}[1]{\underline{\hspace{#1}}}
\begin{document}
        \thispagestyle{empty}
        \begin{center}
                {\LARGE\bfseries CERTIFICATE}
        \end{center}
        \vspace{1cm}

        This is to certify that Mr./Ms. \blank{5cm}, Roll No. \blank{3cm}, a student of M.Sc.-I, Department of
    Mathematics, Mumbai University, has successfully completed the project report titled ``\blank{7cm}'' during the
    academic year \blank{3cm}, under my guidance and supervision.

        \vspace{3cm}
        \noindent Date: \blank{3cm}\hfill Project Guide: \blank{4cm}

        \vspace{1.5cm}
        \noindent Place: \blank{3cm}\hfill Head of the Department: \blank{4cm}
\end{document}
```

Q.19

```
\documentclass{article}
\usepackage{tikz}
\begin{document}
        \begin{tikzpicture}[level distance=15mm,
                level 1/.style={sibling distance=40mm},
                level 2/.style={sibling distance=20mm}]
                \node {India}
                child {node {Maharashtra}
                        child {node {Mumbai}}
                        child {node {Pune}}}
                child {node {West Bengal}
                        child {node {Kolkotta}}
                        child {node {dum dum}}};
        \end{tikzpicture}
\end{document}
```

Q.20

```
\documentclass{article}
\usepackage{colortbl,xcolor}
\begin{document}
        \begin{center}
                \begin{tabular}{>{\columncolor{blue}\color{white}\bfseries}lr}
                        \rowcolor[gray]{0.8}
                        \color{black} Day & \bfseries Attendance\\[2pt]
                        Monday & 57\\ Tuesday & 11\\ Wednesday & 96\\ Thursday & 122\\
                        Friday & 210\\ Saturday & 198\\ Sunday & 40\\
                        \cellcolor[gray]{0.8}\color{black}Total & 724
                \end{tabular}
        \end{center}
\end{document}
```